

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 1: Real Numbers

Previous year board questions, topic-wise: 8 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 HCF and LCM by prime factorisation

1. Let m be the HCF and n the LCM of 98 and 28. Find the value of $n - 7m$. **CBSE 2025, 1 mark**
2. Two positive integers are $p = 18a^2b^4$ and $q = 20a^3b^2$, where a and b are primes. What is the LCM of p and q ? (a) $2a^2b^2$ (b) $180a^2b^2$ (c) $12a^2b^2$ (d) $180a^3b^4$ **CBSE 2024, 1 mark**
3. At a workshop there are 48 French teachers, 80 Hindi teachers and 144 English teachers. Every room must hold the same number of teachers, and all the teachers in a room must teach the same subject. What is the least number of rooms needed? **CBSE 2024, 3 marks**

2 The Fundamental Theorem of Arithmetic

4. For which natural numbers n is $(-1)^n + (-1)^8 = 0$? **CBSE 2025, 1 mark**
5. Show that $5 \times 11 \times 17 + 3 \times 11$ is a composite number. **CBSE 2024, 2 marks**

3 Irrational numbers and their proofs

6. Which of these is a rational number lying between $\sqrt{3}$ and $\sqrt{5}$? (a) 1.4142387954012... (b) $2.32\overline{6}$ (c) π (d) 1.857142 **CBSE 2025, 1 mark**
7. Show that $\frac{1}{\sqrt{5}}$ is an irrational number. **CBSE 2025, 3 marks**
8. Given that $\sqrt{3}$ is irrational, prove that $5 - 2\sqrt{3}$ is irrational. **CBSE 2024, 2 marks**

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